Technology and Artificial Intelligence (INF131)

What you will learn this semester: from data to decisions with AI



Data and spreadsheets
Weeks 1–5

- The digital economy
- Data fundamentals
- Computational thinking
- Cleaning data with Power Query
- Analysis and visualisation

Tools Excel · Power Query

Databases and code
Weeks 6–7

- Databases and SQL
- Programming basics with Python

Tools SQL · Google Colab

Systems and security
Weeks 9–11

- Internet and cloud
- Cybersecurity and data protection
- Enterprise systems and BI

Tools Power BI

AI and digital finance
Weeks 12–14

- How AI works
- Generative AI in practice
- Fintech and AI regulation

Tools AI assistants

Semester project (weeks 4–15): answer an economic question with real data, raw data to dashboard

Assessment: 40% coursework · 60% exam

Why study Technology and Artificial Intelligence?

- **Employers expect these skills.** About 92% of jobs require digital skills, and jobs needing several of them pay up to 45% more.
- **You learn the tools analysts use every day:** Excel, Power Query, SQL, Power BI and Python, all practised hands-on with real economic data.
- **You learn to use AI well:** how it works, where it makes mistakes, and how to check its output.
- **You understand the systems behind business and finance:** databases, cloud, cybersecurity, enterprise systems and the technologies changing banking.
- **You build a foundation and a portfolio.** The course prepares you for later courses, and your semester project gives you a real dashboard and report to show employers.

Today's plan

Part 1
50 min

1. Why information systems matter
2. What an information system is
3. IoT, big data, cloud and AI
4. Complementary assets

Part 2
50 min

1. Business processes
2. Systems for each management level
3. Enterprise applications
4. Collaboration and knowledge

Each part ends with a case study for discussion. Short questions for you at the end.

By the end of today you will be able to

- Explain why information systems are essential to business today
- Define an information system and tell data from information
- Analyse a system by its organization, management and technology
- Explain why IT pays off only with complementary assets
- Match TPS, MIS, DSS, ESS, ERP, SCM and CRM to the needs they serve
- Choose collaboration tools with the time/space matrix

PART 1

Information systems in global business today

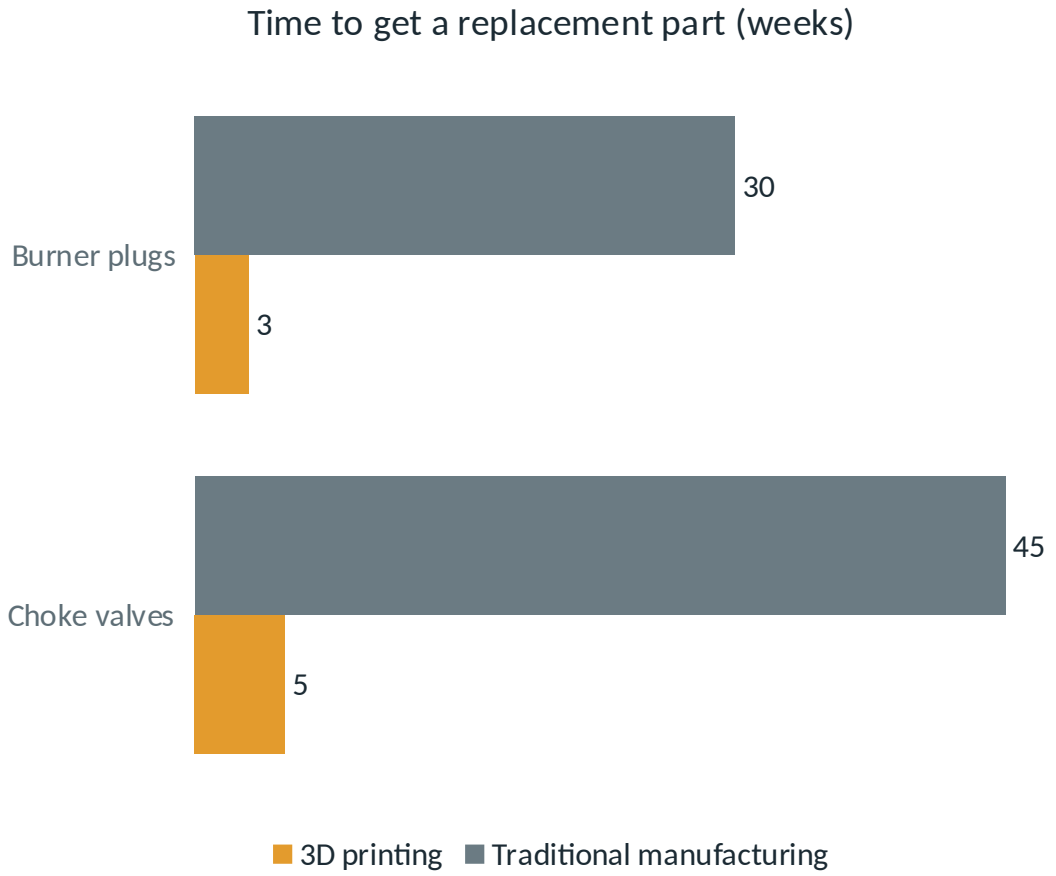


Laudon & Laudon, Chapter 1

ConocoPhillips cut spare-part lead times from months to weeks

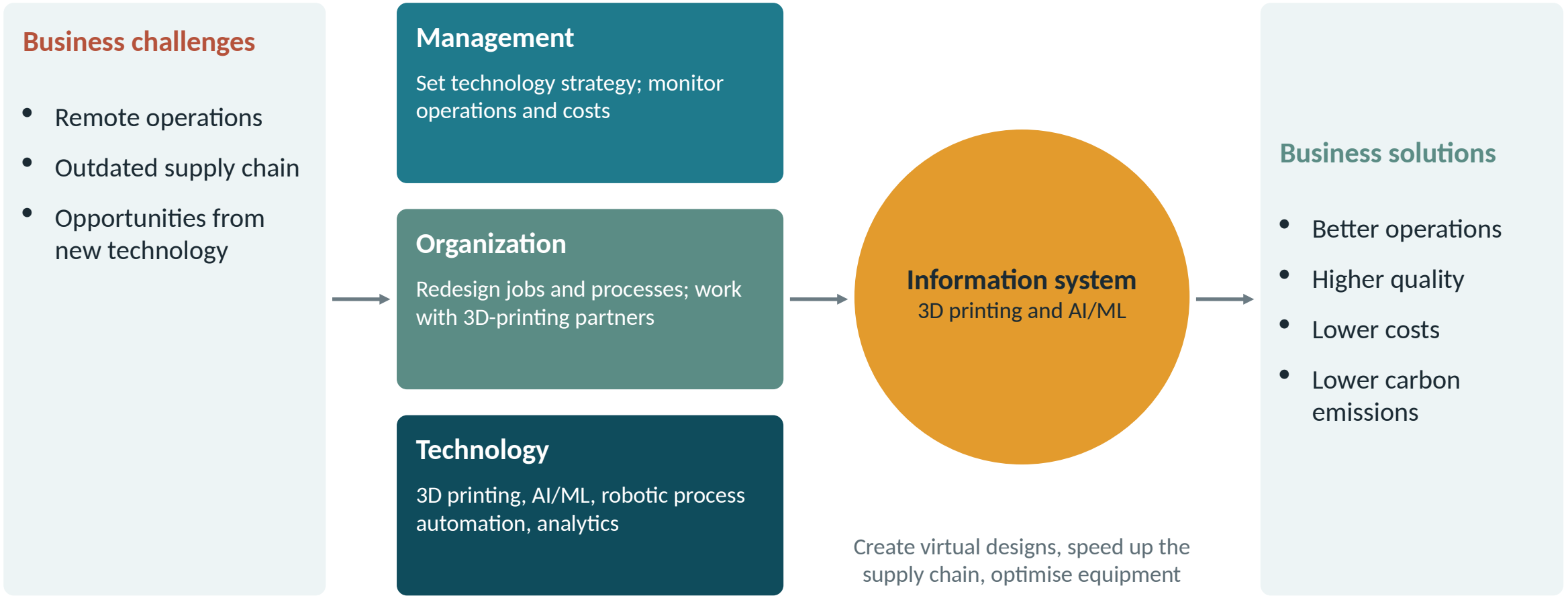
Opening case · Kuparuk oil field, Alaska

- Remote oil field; key turbine parts no longer sold commercially
- Local machine shops needed 30+ weeks to make each part, so spare parts sat in costly warehouses
- Solution: 3D-print parts from digital designs, close to where they are needed
- Also uses data analytics, digital twins, robotic process automation and AI/ML
- Jobs and business processes had to be redesigned as well



Source: Laudon & Laudon (2026), Ch. 1 opening case. 3D-printing times are upper bounds (2–3 weeks; 5 weeks or less).

Every case in the book follows the same analysis framework



Adapted from the opening case diagram, Laudon & Laudon (2026), Ch. 1.

Information systems now shape how markets work



2.9 bn

people expected to buy something online in 2024



~4 bn

social network users, about half the world



$900 bn+

digital ad spending by 2027, over 75% of all media ads



330 m TB

of new data created every day (estimate)

The newest wave is AI: used well, it lowers costs, reduces risk, speeds time to market and creates better products.

Source: Laudon & Laudon (2026), Section 1.1, citing eMarketer and other estimates.

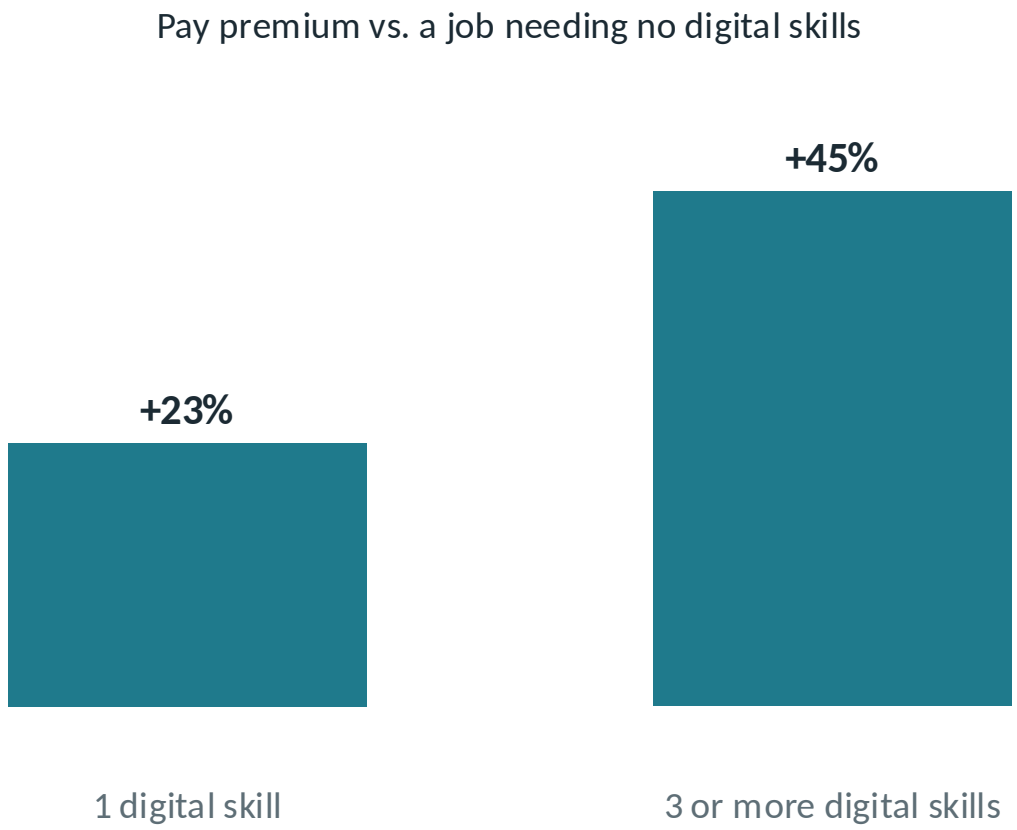
Digital skills pay off in the job market

92%

of jobs require digital skills

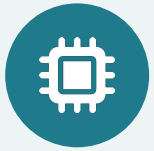
95%

of entry-level positions require them too



Source: National Skills Coalition (2023), analysis of 43 million U.S. job postings, cited in Laudon & Laudon (2026), Fig. 1.1.

New technology creates challenges at three levels



Technology

- Cloud computing
- Big data and the Internet of Things
- Artificial intelligence
- The mobile platform

How much of our IT should move to the cloud?



Management

- Return on IT investment
- Online collaboration and social networks

How do we measure what our IT spending returns?



Organizations

- Security and privacy
- Social business
- Remote and hybrid work

How do we know our data are secure?

Technology made globalization possible, with winners and losers

Globalization

The flow of goods, services, capital, people and ideas across international borders.

~60%

world trade as a share of global economic activity

>35%

of U.S. Fortune 500 revenue earned abroad

Opportunities

- New markets with billions of customers
- Low-cost suppliers and production abroad
- Near-free, instant global communication
- Price and quality information 24/7

Challenges

- Stronger competition from global rivals
- Jobs, including skilled ones, move offshore
- You need skills that cannot be done cheaper abroad (Your competitor is not your classmate anymore)

Source: Laudon & Laudon (2026), Section 1.2, citing World Trade Organization (2023).

A digital firm runs its key relationships and assets digitally



Digital relationships

Customers, suppliers and employees are connected digitally



Networked processes

Core business processes run over digital networks



Digital assets

Intellectual property, competencies, finances and people are managed digitally



Fast response

Senses and responds to change quickly; works 24/7 (time shifting) and globally (space shifting)

Digital Firm: Company in which the organization's significant business relationships with customers, suppliers, and employees are digitally enabled and corporate assets are managed through digital means.

Digital transformation: large-scale change to capture the benefits of digital technology. It is ongoing and needs new business models, structures and a data-driven culture, not just new tools. Examples: Cisco, Caterpillar, 3M.

Firms invest in information systems for six strategic objectives

Operational excellence

Walmart's Retail Link connects suppliers to every store, Replenishment System to replace the shelf; $648 bn sales in fiscal 2024

New products and business models

Business Model: How a company produces, delivers, and sells a product or service to generate revenue.
Apple turned music from CDs into digital downloads with iTunes

Customer and supplier intimacy

Ritz-Carlton tailors each stay to the guest's profile based on previous data;
TAL Apparel restocks shirts from live sales data

Better decision making

Data-Driven Decision making:
Coca-Cola Consolidated sales teams use real-time dashboards in the field

Competitive advantage

Reaching the first four objectives lets firms like Apple, Walmart and UPS outperform rivals

Survival

ATMs became a must for every bank; laws such as Sarbanes–Oxley require digital record keeping

New expectations: sustainability, governance and responsible AI



Sustainability

Energy use, carbon footprint, recycling, pollution, resource conservation



Governance

Leadership conduct, executive pay, internal controls, shareholder rights



Responsible AI

Guidelines so AI is used ethically, fairly and transparently

Example: Crowley

Logistics firm that measures its own and its suppliers' emissions with Salesforce Net Zero Cloud.

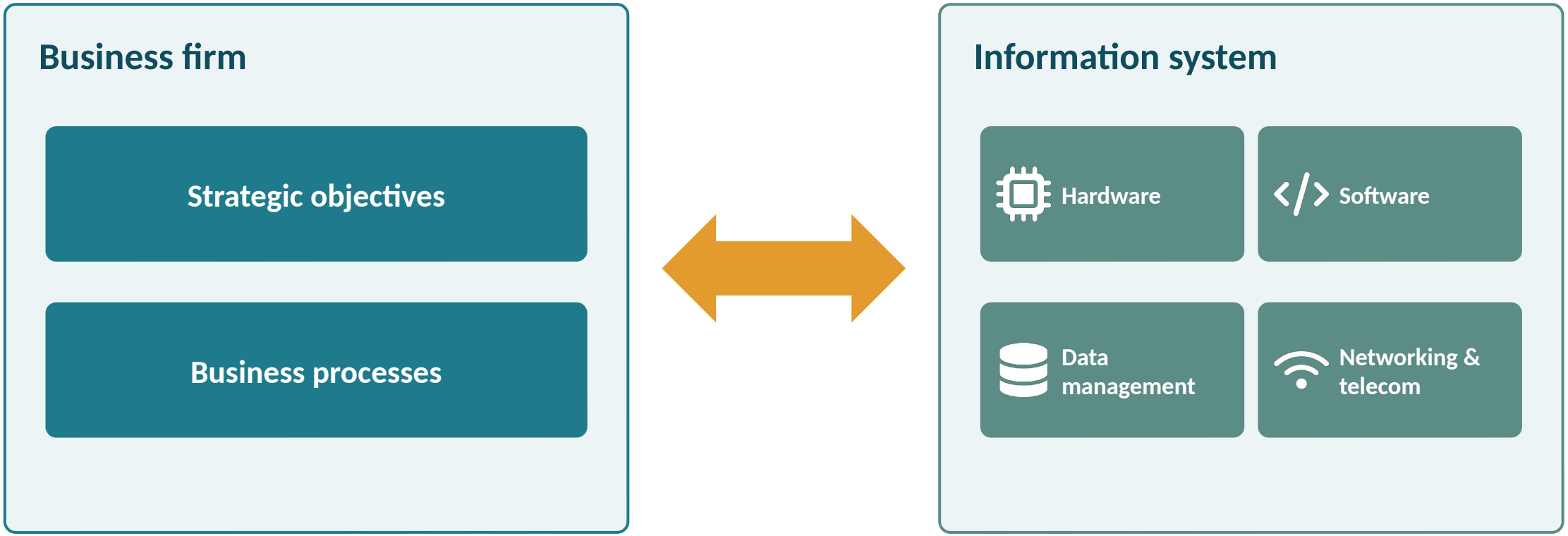
4.2 m

metric tons of greenhouse gases expected to be cut per year

≈ removing 900,000 cars from the road

Source: Laudon & Laudon (2026), Section 1.4, citing PwC (2022).

Strategy and information systems depend on each other



What a firm wants to do in five years often depends on what its systems will be able to do.

Information system (IS): A set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision-making, coordinating, and control in an organization. In addition, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products.

Data become information when they are organized to be useful

Data: raw facts

```
331 Brite Dish Soap   1.29
863 BL Hill Coffee    4.69
173 Meow Cat          0.79
331 Brite Dish Soap   1.29
663 Country Ham       3.29
524 Fiery Mustard     1.49
331 Brite Dish Soap   1.29
...
```

Information system: total, classify, analyse

Information: meaningful

Superstore #122 · Northwest region

Brite Dish Soap (item 331)

7,156

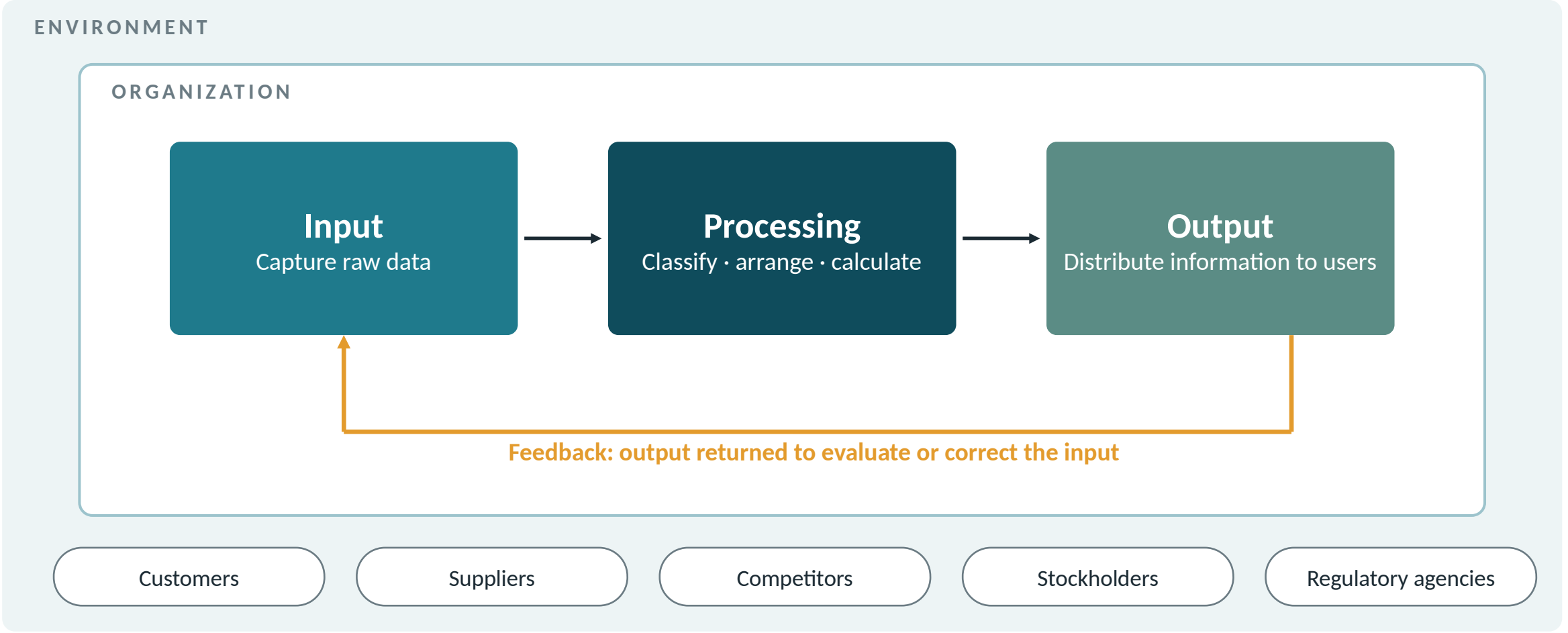
units sold

$9,231.24

year-to-date sales

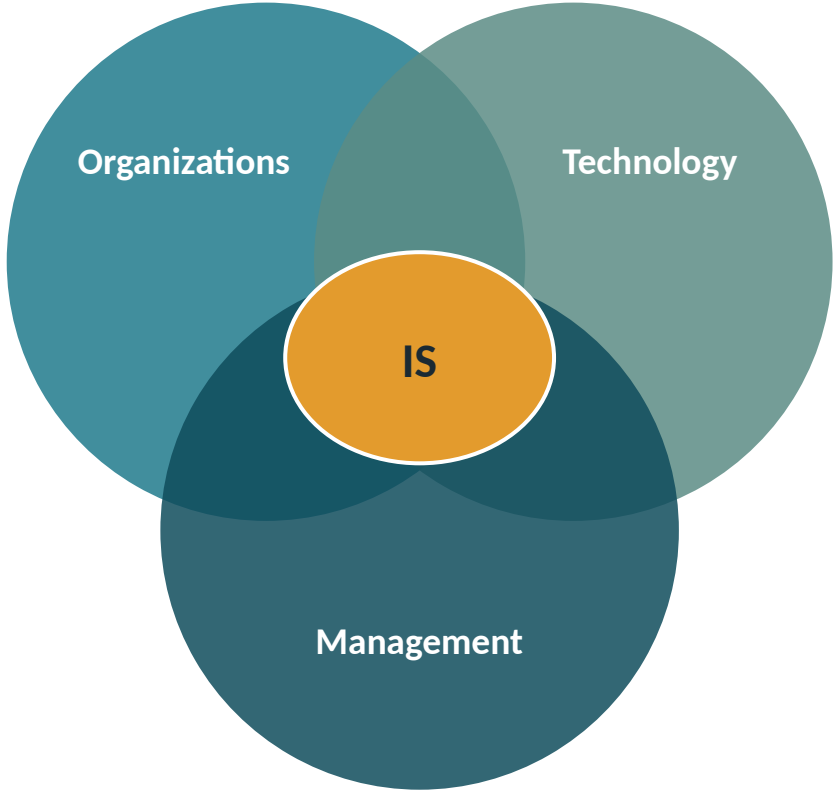
Example from Laudon & Laudon (2026), Figure 1.3: supermarket checkout data.

An information system turns input into output, with feedback



Adapted from Laudon & Laudon (2026), Figure 1.4.

Information systems are more than computers



The house analogy. Computers and software are the hammer, nails and lumber. They do not make a house without architecture, design and decisions.

Computer literacy

Knowing how IT and computers work

Information systems literacy

Also understanding the organizations, managers and people who use systems: the goal of this course

Adapted from Laudon & Laudon (2026), Figure 1.5.

The organization dimension: levels, functions, processes, culture



Business functions

Sales & marketing · manufacturing & production · finance & accounting · human resources

Business processes

Formal and informal ways of getting work done, e.g., billing a customer

Culture

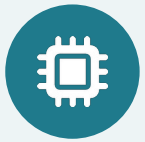
Shared values, built into systems: UPS's customer focus shows in package tracking

Politics

Different levels and functions have different interests; systems come out of compromise between interested/conflicting parties

Based on Laudon & Laudon (2026), Figure 1.6 and Table 1.2.

The technology dimension: four components, one platform



Hardware

Physical equipment for input, processing and output



Software

Instructions that control the hardware



Data management

Software that organises data on storage



Networking & telecom

Devices and software that link computers and move data

Internet Global network of networks; 4.8 bn users (~60% of the world)

Intranet Internal company network using Internet standards

Extranet Intranet opened to authorized outsiders, e.g., suppliers

IT infrastructure = all of these plus the people who run them: the shared platform on which every system is built.

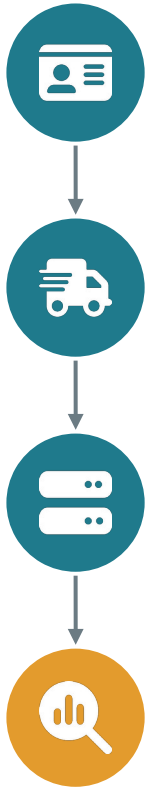
Case Study: UPS information systems

25 m
packages delivered per day

743.5 m
tracking requests per day

$1 bn
spent on IT every year

100 m
miles saved per year by ORION route optimisation



Smart label
Bar-coded label holds sender, destination, delivery time

DIAD handheld
Driver's device downloads the route, captures signatures

Central computers
Tracking data stored and processed

Customers track
Website, app and service reps see status in real time

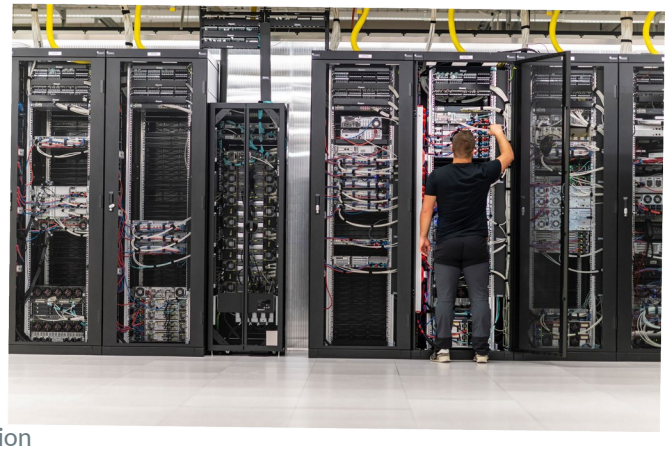
AI at UPS: ORION routing · DeliveryDefense · storm rerouting (NPT)

Source: Laudon & Laudon (2026), Ch. 1 Spotlight on Technology case. ORION also saves ~10 million gallons of fuel per year.

The Big Four: IoT, big data, cloud and AI

- Connected cars
- Appliances
- Sensors
- Medical devices
- Machine tools

Cloud computing
Shared storage and processing power over the Internet



Big data
Huge volume, variety and velocity

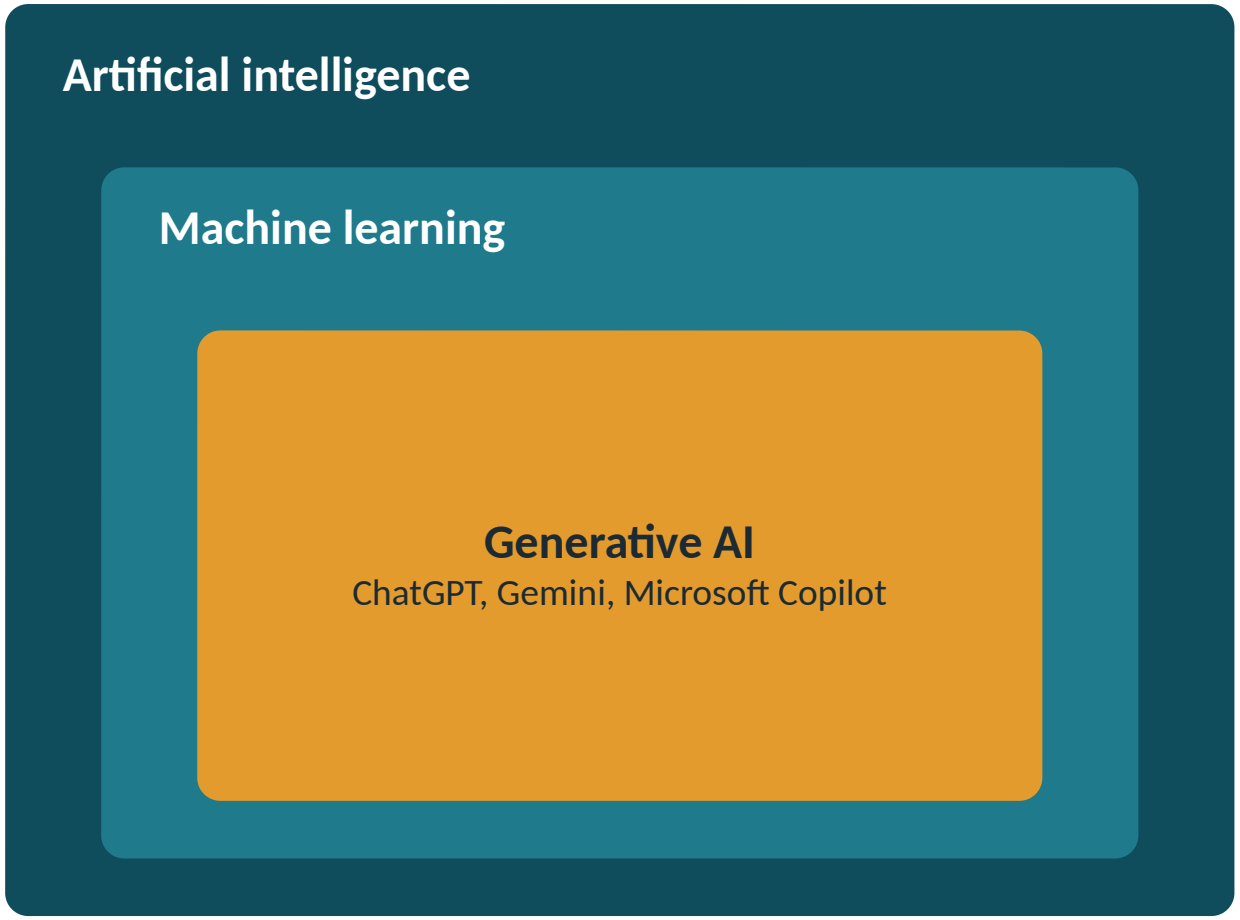
Analytics and AI
Find patterns, predict, generate content

Results
Spot cancer on X-rays · predict machine failure · guide a car

Internet of Things: ~22 bn connected devices by 2025

Adapted from Laudon & Laudon (2026), Figure 1.7.

AI, machine learning and generative AI are nested ideas



Artificial intelligence

Computer systems that do tasks normally needing human intelligence: recognising speech and images, spotting patterns, predicting, learning

Machine learning

AI that learns and improves on its own by finding patterns in very large amounts of data

Generative AI

Tools that learn patterns in existing data to create new text, code, images and music

Definitions: Laudon & Laudon (2026), Section 1.7. AI gets its own chapter later (Chapter 11).

IT pays off only with Complementary Assets

Complementary assets are those assets required to derive value from a primary investment.



Same spending, very different results. The difference: changes in organization and management.

Organizational

Supportive culture · right business model · efficient processes · decentralized decisions · strong IS team

Managerial

Senior management support · incentives to innovate · teamwork · training · flexible culture

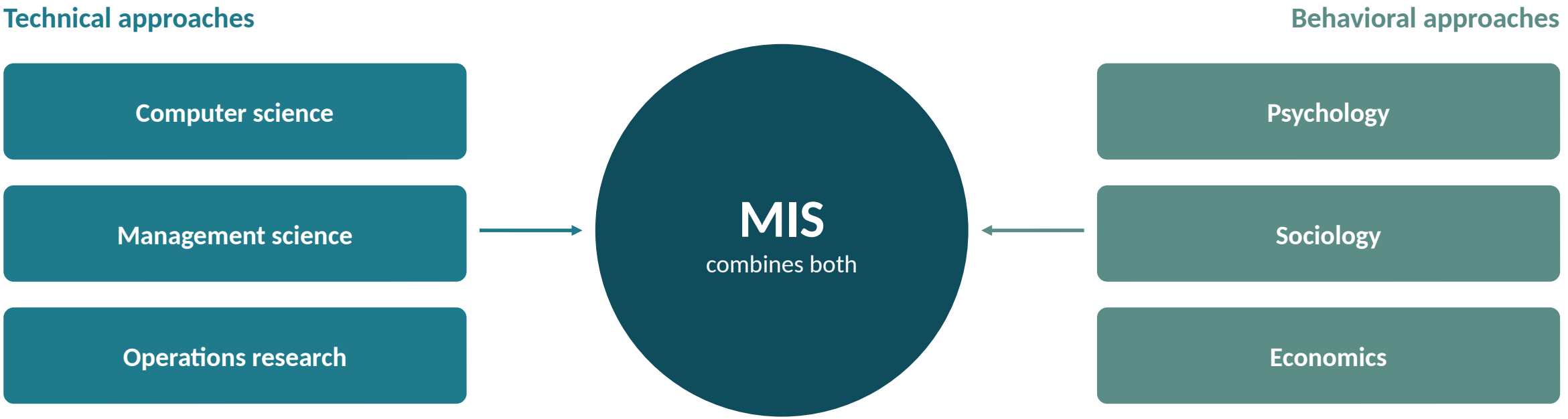
Social

Internet infrastructure · education · standards · fair laws · technology service firms nearby

Analogy: cars create value only with roads, petrol stations, repair shops and traffic rules, otherwise you get a living hell like Tirana!

Based on Laudon & Laudon (2026), Section 1.8 and Table 1.3.

Studying information systems needs a sociotechnical view



Sociotechnical view: best performance comes when technology and organization adjust to each other until they fit. A new system that people do not use, or new processes without systems to support them, add nothing.

Information Systems (IS) is a broad umbrella term for any system that collects, processes, and stores data, while a **Management Information System (MIS)** is a specific type of IS designed to help managers make strategic business decisions.

Based on Laudon & Laudon (2026), Figures 1.8 and 1.9.

PART 2

Global e-business and collaboration



Laudon & Laudon, Chapter 2

Toyota North America: Teams and generative AI at work

Opening case · Toyota Motor North America (TMNA)

- 14 plants, 1,500 dealerships, 49,000+ employees
- Covid-19 closed plants and offices; Microsoft Teams was rolled out to the whole workforce
- Executives inspected factory floors through daily virtual meetings
- AgentAsk, a ChatGPT-like assistant inside Teams, answers IT, finance, legal and travel questions

11.4 min

average time to fix an IT issue with AgentAsk

vs

3 days

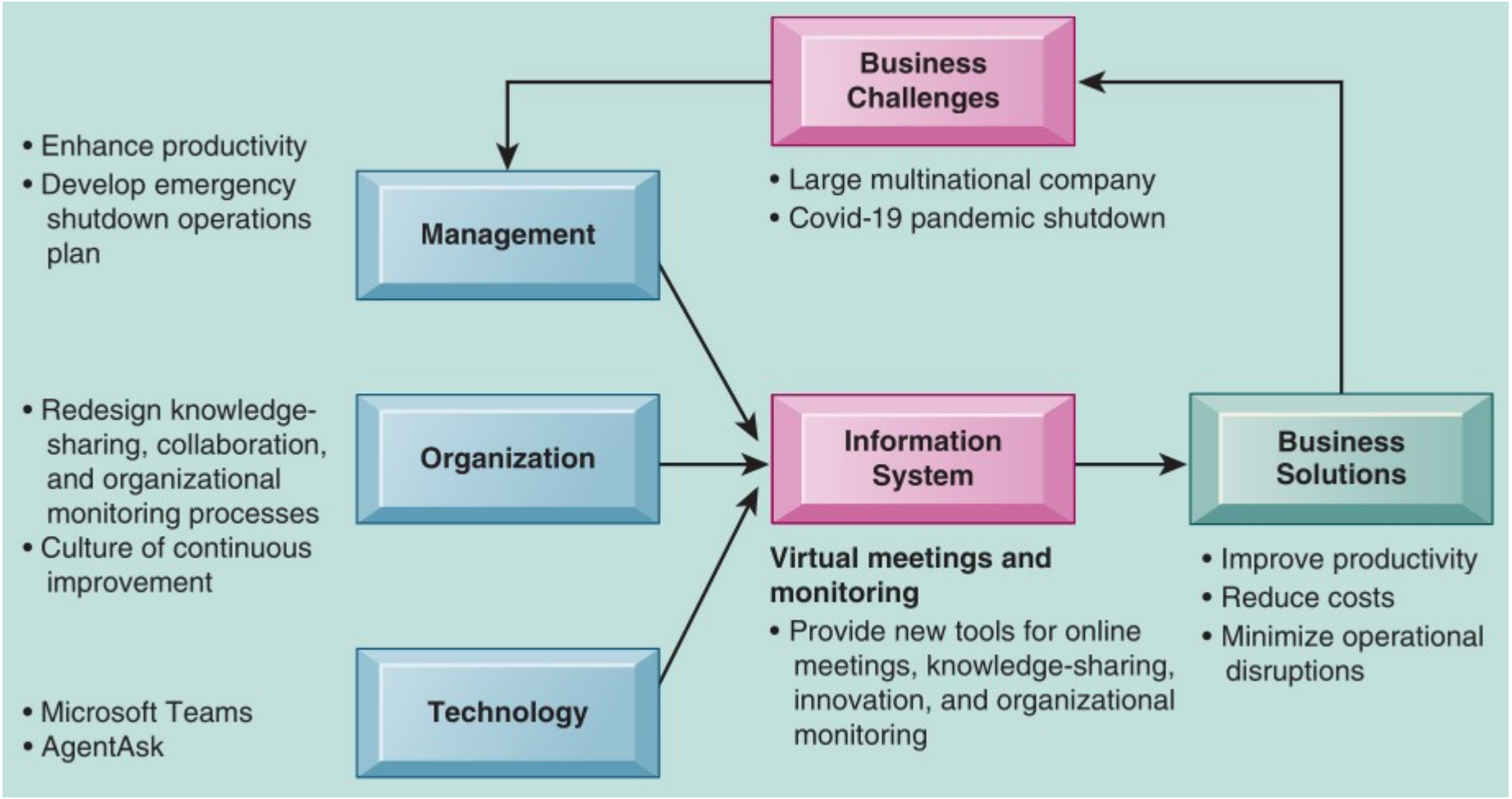
industry average

70,000+

hours of productivity given back to employees and service desk agents in one year

Source: Laudon & Laudon (2026), Ch. 2 opening case.

Toyota North America: Teams and generative AI at work



A business process is how work gets done

Business process: the set of activities, and flows of material, information and knowledge, needed to produce a product or service. Good processes are a strength; bad ones are a liability. Information systems make it possible for firms to manage all their information, make better decisions, and improve the execution of their business processes.

Manufacturing and production

- Assemble the product
- Check quality
- Produce bills of materials

Sales and marketing

- Identify customers
- Make customers aware
- Sell the product

Finance and accounting

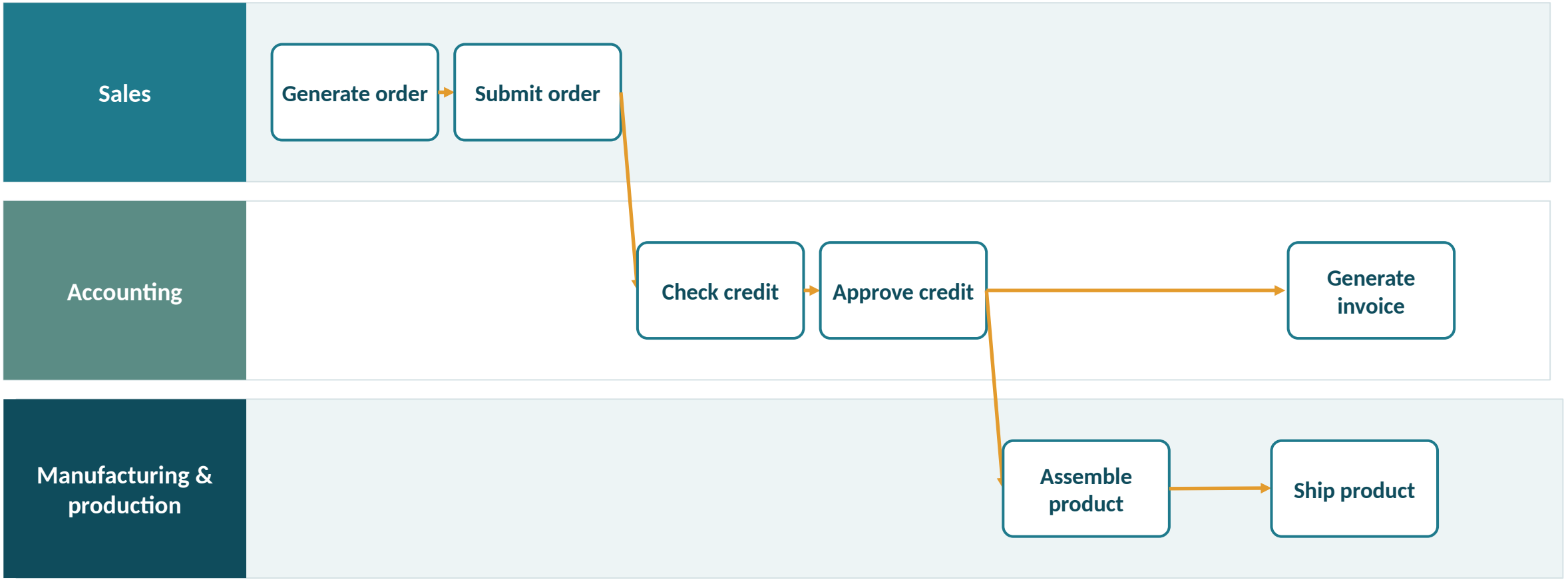
- Pay creditors
- Create financial statements
- Manage cash accounts

Human resources

- Hire employees
- Evaluate performance
- Enrol in benefits plans

Based on Laudon & Laudon (2026), Table 2.1.

Fulfilling one order crosses three departments



The process is cross-functional and also interorganizational: delivery firms and customers take part. Each step needs fast information flow.

How information technology improves business processes

Automate	Share	Speed up	Transform
Do manual steps automatically, e.g., credit checks and invoices	Let many people see the same information at once	Run steps in parallel instead of in sequence; remove decision delays	Enable new business models, e.g., Amazon, Apple Music



Why analyse processes? Mapping a process shows how a business really works, and where technology can make it faster, cheaper or better for customers. It also means you can create your IT product and sell it → $$$

Each management level needs a different kind of system

Transaction Processing System (TPS): a software-based information system that collects, stores, modifies, and retrieves the routine, everyday data transactions of an enterprise.



ESS (Executive Support Systems) · strategic, non-routine

"What products should we be making in five years?"

DSS (Decision-Support Systems) · modeling, unique, fast-changing

"What happens to production if December sales double?"

MIS (Management Information Systems) · routine reports, descriptive

"Are we meeting this year's sales targets?"

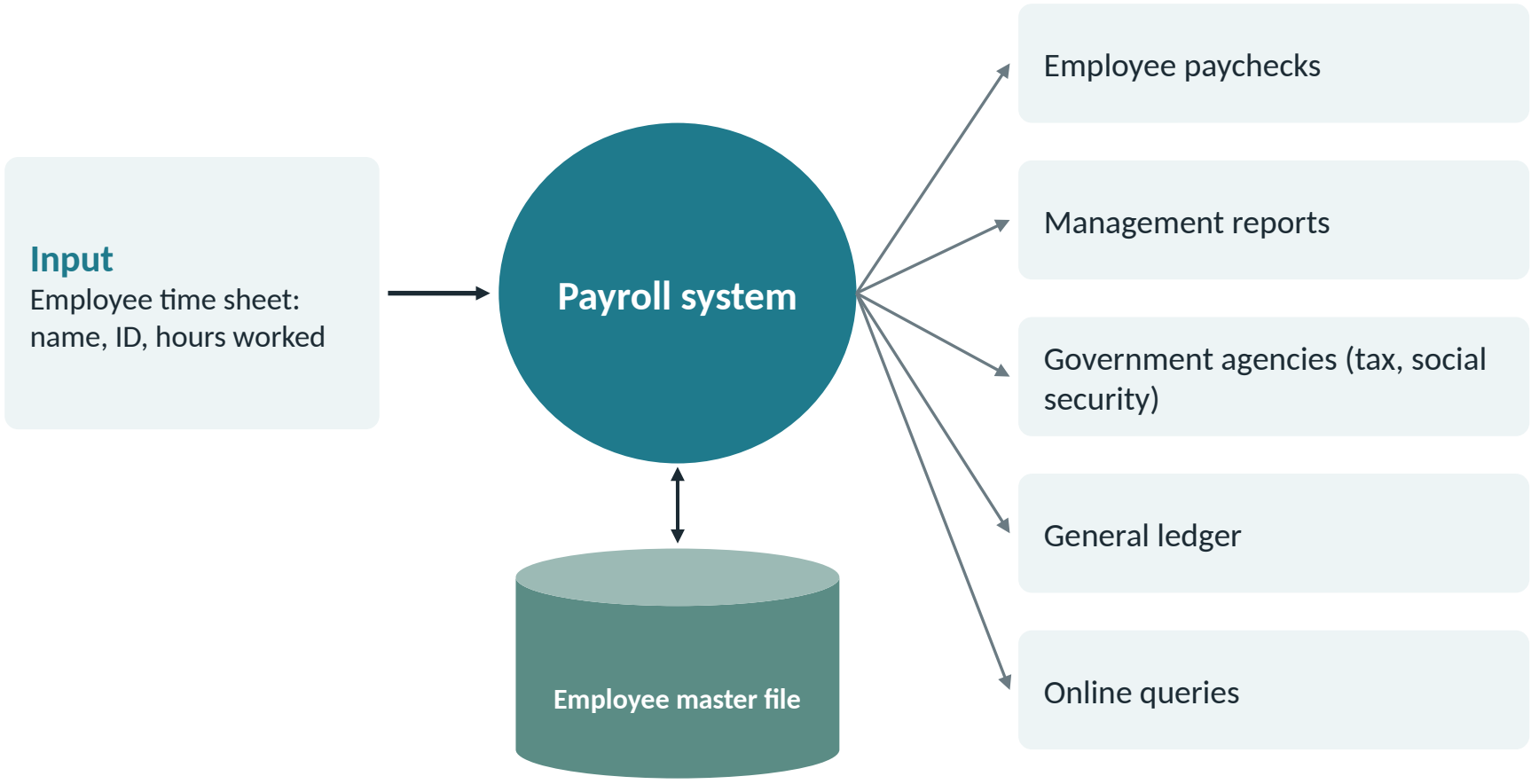
TPS · daily transactions

"How many parts are in inventory?" "What happened to Mr. Garcia's payment?"

MIS, DSS and ESS together = business intelligence (BI): tools that help managers make better decisions

***Business intelligence (BI)** is a contemporary term for data analytics and other software tools for organizing, analyzing, and providing access to data to help managers and other enterprise users make more informed and thus better business decisions.*

Transaction processing systems keep the business running



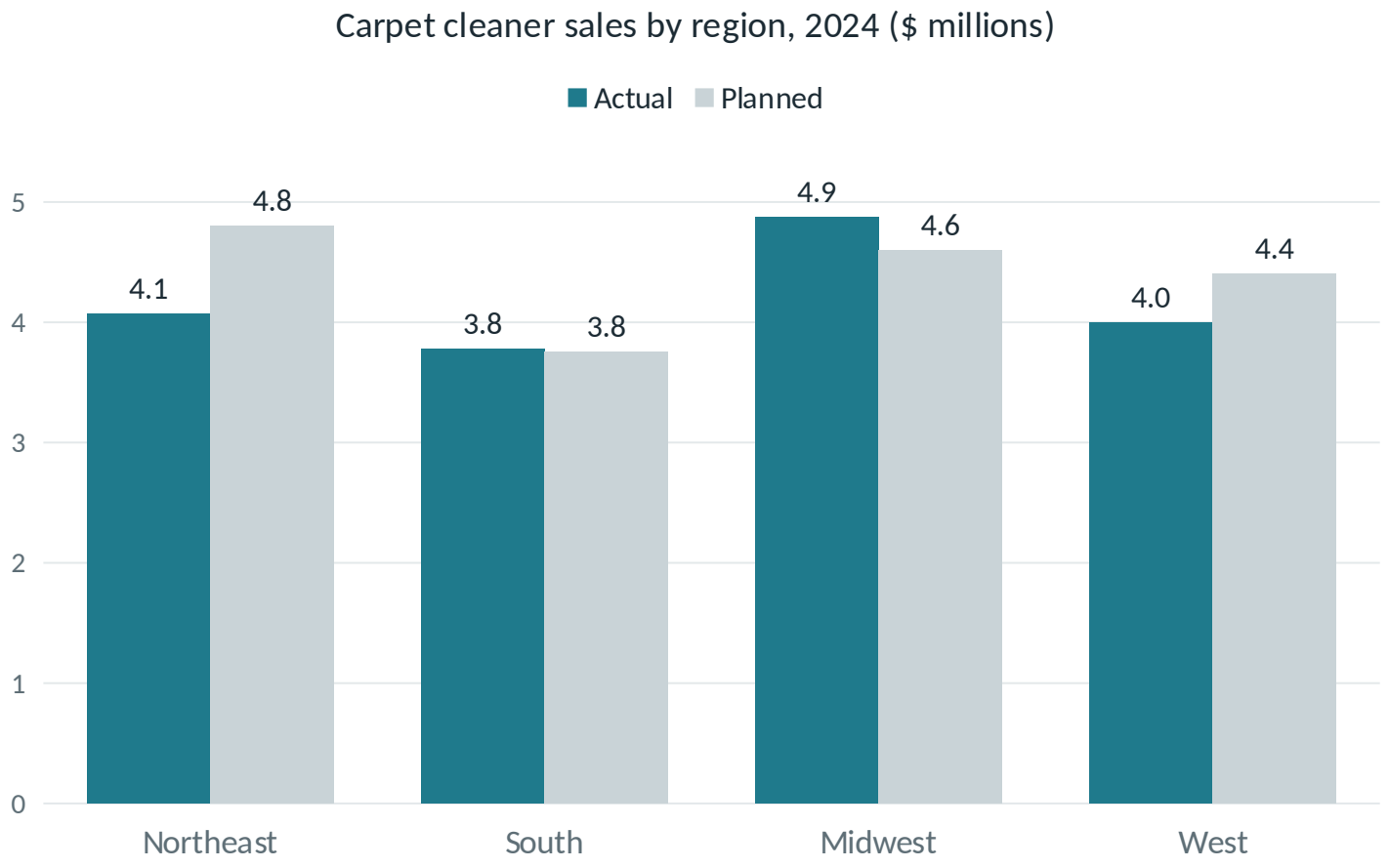
When a TPS fails

~1,700

Southwest Airlines flights affected when a computer failure grounded all its flights on the morning of 18 April 2023

Adapted from Laudon & Laudon (2026), Figure 2.2 and Section 2.2.

An MIS report compares actual results with the plan



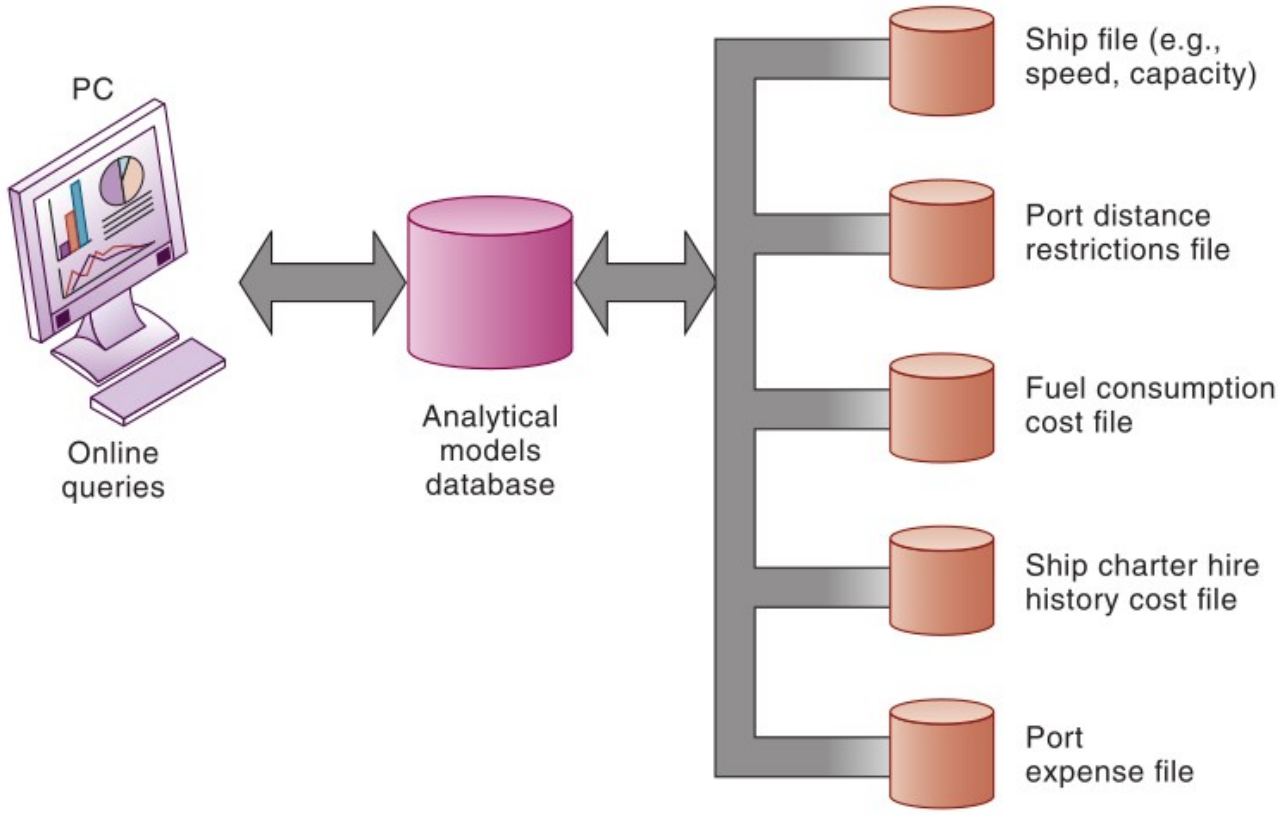
Actual ÷ planned	
Northeast	0.85
South	1.01
Midwest	1.06
West	0.91
Total	0.95

Typical MIS

- Built from TPS data
- Routine questions, fixed schedule
- Summaries and comparisons, little analysis

Data: sample MIS report for a fictional company, Laudon & Laudon (2026), Figure 2.4.

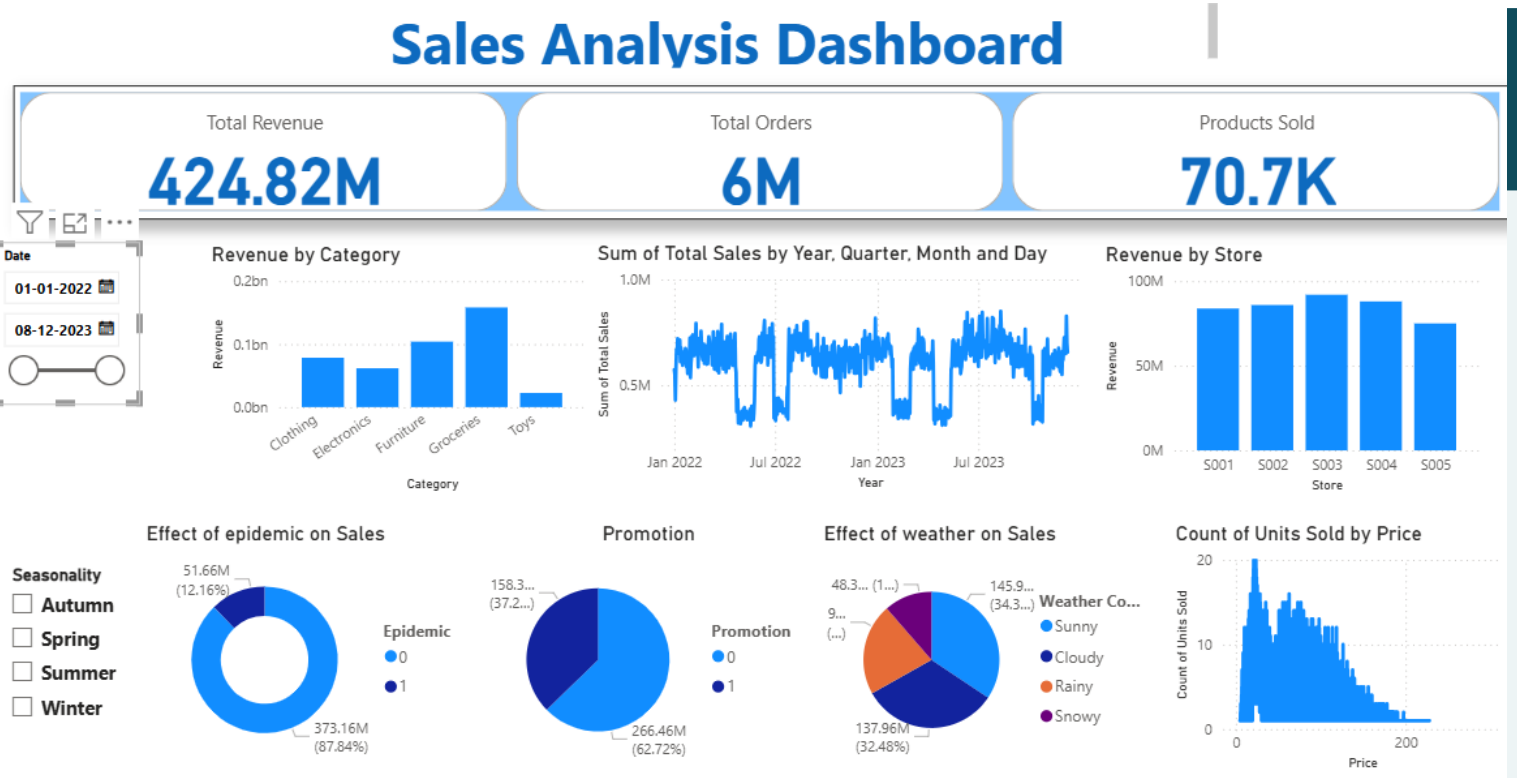
A Complicated DSS Schema for shipping



Decision-support system (DSS)

Users	Middle managers, business analysts
Decisions	Unique, fast-changing, "what-if"
Uses	Data plus analytical models, often external data. Heavily model-based and depending on the problem may require lots of compute power
Example	Shipping firm's voyage-estimating DSS: which ship, at what speed and rate, earns the most?

ESS support harder, non-routine decisions



Executive support system (ESS)

Users	Senior managers
Decisions	Strategic, non-routine, long-term trends
Uses	Summaries of internal and external data via portals and dashboards
Example	Sales analysis on the right: Total Revenue, Total Orders, Sales, Times, Seasons, Trends ...

Based on Laudon & Laudon (2026), Section 2.2 and Figure 2.5.

Enterprise applications connect the whole firm and its partners



What happens when you buy an iPhone from Amazon?



Adapted from Laudon & Laudon (2026), Figure 2.6 and Section 2.3. Covered in depth in Chapter 9.

Collaboration matters more than ever

Collaboration: working with others to achieve shared and explicit goals, from a few minutes to years, one-to-one or many-to-many.

Work has changed

Jobs need close coordination instead of isolated silos

More professional work

Experts must share knowledge to get work done

Flatter firms

Teams, not orders from the top, decide how work is done

Global scope

Ford: 177,000 employees in 65+ plants worldwide (2024)

Innovation is a team effort

Even famous innovators relied on strong teams

Culture of teamwork

Diverse teams and the "wisdom of crowds"

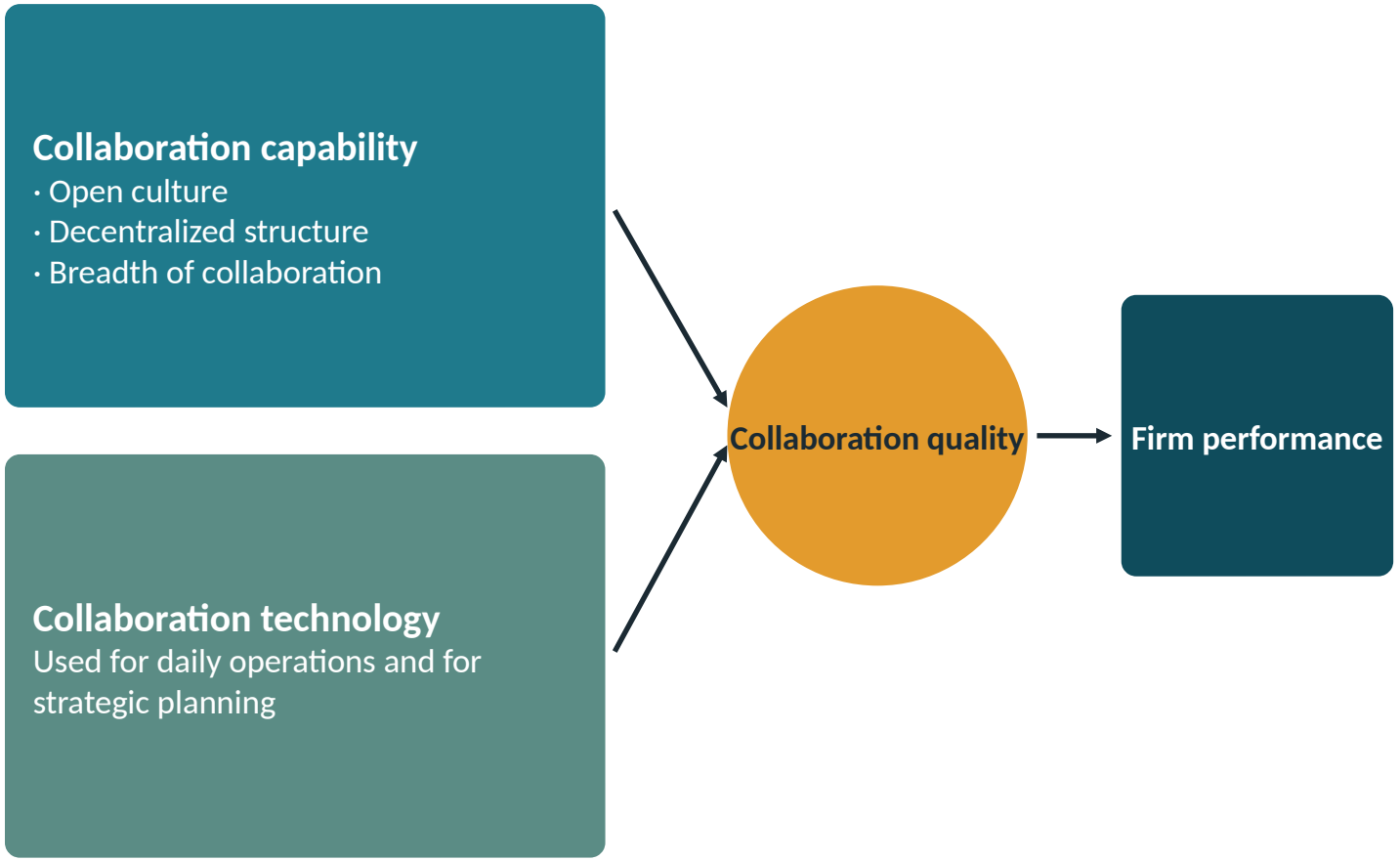
Collaboration pays, but only with the right culture

5×

more likely to be high-performing: firms that promote collaboration

But…

Two-thirds of the 1,100 firms studied called collaboration a core value; only a minority got good results.



Sources: Institute for Corporate Productivity and Rob Cross study, cited in Laudon & Laudon (2026), Section 2.4; diagram adapted from Figure 2.7.

The collaboration toolbox

*A collaborative, team-oriented culture won't produce benefits without **information systems** in place to enable collaboration, group decision making, and social business.*



Messaging

Slack, Teams chat, instant messaging (IM)



Virtual meetings

Zoom, Microsoft Teams, telepresence



Cloud file sharing and co-editing

Google Drive and Docs, OneDrive, Dropbox



Team sites and documents

Microsoft SharePoint: workflows, versions, wikis, search



Enterprise social networks

Microsoft Viva Engage, Salesforce Chatter



Virtual worlds and metaverse

KPMG's metaverse collaboration hub

Humans and AI increasingly collaborate

Human–AI collaboration: people and AI combine their strengths to reach a shared goal, with better results than either alone.

Medicine

AI scans X-rays, MRIs and ECGs for abnormalities; radiologists confirm the diagnosis

Marketing

CarMax used GPT-3.5 to summarise 5,000 customer reviews; staff checked the text before publishing

Finance

JPMorgan traders use a model that read 25 years of Federal Reserve speeches for trading signals

Drug discovery

AI predicts compound properties and toxicity; Merck uses generative AI to design new molecules

Source: Laudon & Laudon (2026), Table 2.4 and Spotlight case "AI Accelerates Drug Discovery and Development".

Choose collaboration tools by time and place

	Same time (synchronous)	Different time (asynchronous)
Same place	**Face-to-face** Decision rooms, shared tables, wall displays, AI tools	**Continuous task** Team rooms, large public displays, project management, AI tools
Different place	**Remote interaction** Videoconferencing, instant messaging, shared screens, multi-user editors	**Communication + coordination** Email, blogs, wikis, group calendars, workflow, version control

Six steps to pick tools

1. Locate your firm in the matrix
2. List products for each cell
3. Compare costs and benefits, including training
4. Check security and vendor risk
5. Ask users about ease of use
6. Shortlist and invite demos

Adapted from Laudon & Laudon (2026), Figure 2.8 and Section 2.7.

Knowledge management: capture what the firm knows

Knowledge management (KM): the set of business processes developed in an organization to create, store, transfer, and apply knowledge.

Explicit knowledge (Structured Knowledge)

Written down: reports, rules, manuals, documents

Tacit knowledge

In employees' heads: experience and know-how, never documented

80%+

of business content is unstructured or semi-structured: emails, memos, chats, slides, videos



Enterprise-wide KMS

Content management systems, collaboration tools, learning management systems



Knowledge work systems

Powerful workstations for engineers and scientists, e.g., computer-aided design

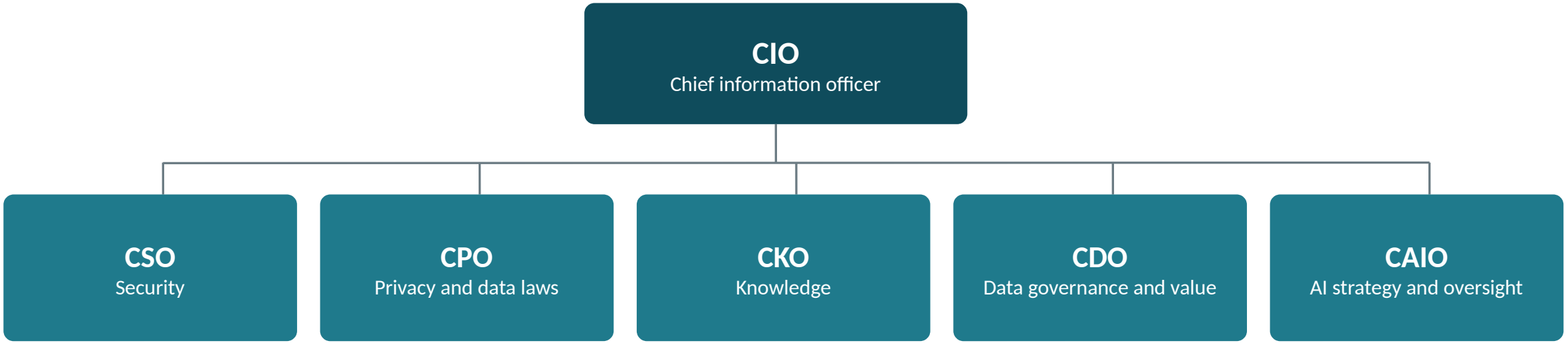


Intelligent techniques

Data mining, machine learning, neural networks, natural language processing, expert systems

Based on Laudon & Laudon (2026), Section 2.8 and Figure 2.9.

Who runs information systems in a firm?



Programmers
Write the software

Systems analysts
Translate business needs into system requirements

IS managers
Lead teams of specialists

End users
Departments the systems are built for; growing role in design

IT governance: who decides about IT, how decisions are made and monitored, so IT supports the firm's strategy.

Based on Laudon & Laudon (2026), Section 2.9.

KEY TAKEAWAYS

Five ideas to remember

1. An information system is organization + management + technology, not just computers
2. Data become information when processed: input → processing → output, with feedback
3. IT pays off only with complementary assets: processes, culture, skills
4. TPS, MIS, DSS and ESS serve different management levels; ERP, SCM and CRM connect them
5. Collaboration tools work only when culture and processes support them